

PACMAN

ARTIFICIAL INTELLIGENCE PROPOSAL

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PACMAN

Pacman is a classic arcade game where the player controls Pacman to navigate a maze, eating dots and avoiding ghosts. The goal is to eat all the dots while surviving and maximizing the score.

Using A* Algorithm.

Pacman is a model-based, goal-based, autonomous agent in a multi-agent environment.



**For Pacman, the agent type is:
model-based,**

**Pacman can remember parts of the maze
(walls, ghosts' movement), not acting
only on immediate perception.**

goal-based,

**Pacman's actions are driven by the goal:
eat all dots and stay alive.**

autonomous agent in a

**Pacman acts on its own after receiving
inputs (from the environment).**

multi-agent environment.

**If you focus only on Pacman, it's a single-
agent environment. But when including
ghosts (as opponents), it's a multi-agent
environment.**



PACMAN Algorithm

A* Algorithm in Pacman:

Find the shortest and smartest path from Pacman's current position to a target (like a dot, fruit, or away from a ghost) in the maze.



AGENT DESIGN (PEAS)

PEAS

**Performance measure,
Environment, Actuators, Sensors**

Performance Measure

Maximize score, minimize lives lost, eat all dots, eat bonus fruits, capture vulnerable ghosts.

Environment

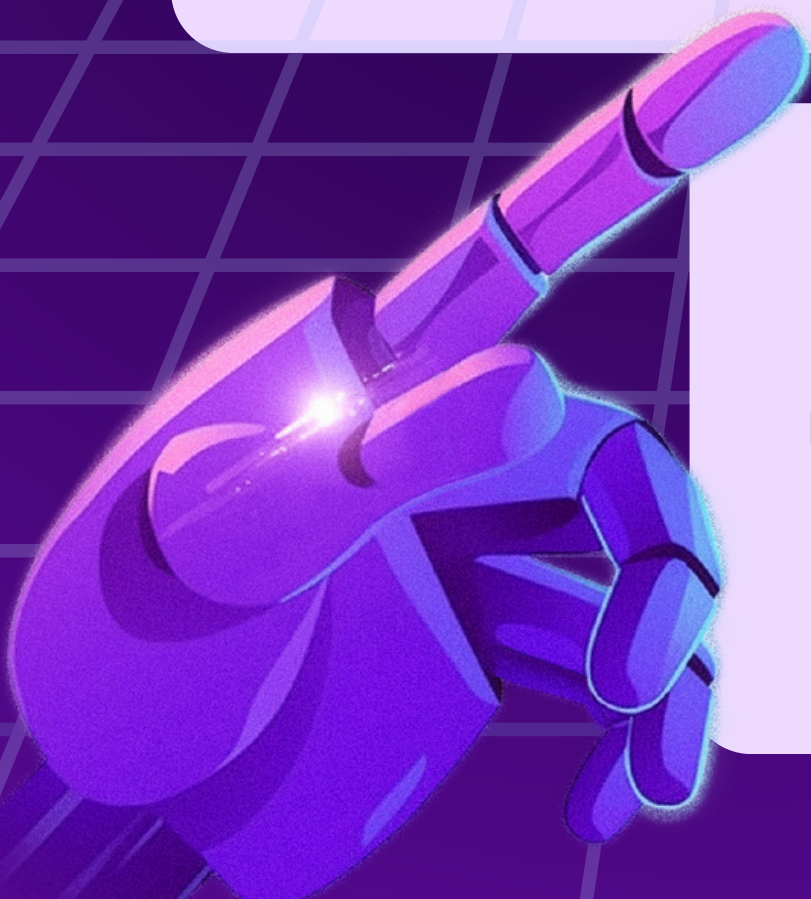
Maze with walls, dots, power pellets, ghosts, fruits.

Actuators

Movement in four directions (up, down, left, right).

Sensors

Detect nearby walls, dots, ghosts, power pellets, fruits.



ENVIRONMENT PROPERTIES (ODESDA)

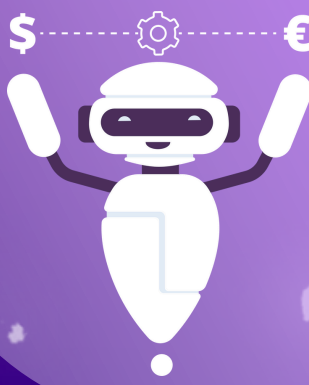
OBSERVABLE

Partially Observable (Pacman can only "see" part of the maze around him).



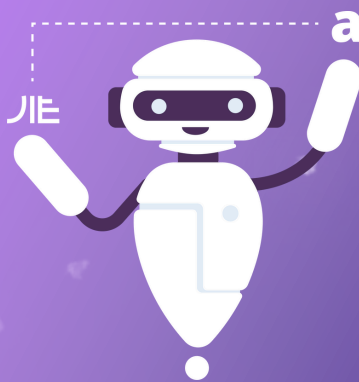
DETERMINISTIC

No, it's Stochastic because ghosts move unpredictably.



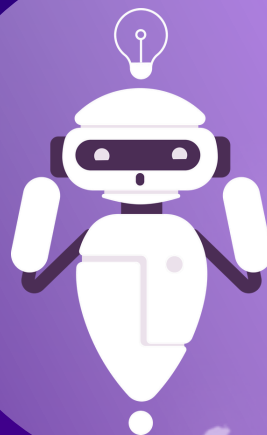
EPISODIC

No, it's Sequential — each action affects future situations.



STATIC

No, it's Dynamic — ghosts move even if Pacman doesn't.



DISCRETE

Yes, the maze, dots, and movements are discrete (grid-based).



AGENTS

Yes — Pacman (player) and Ghosts (enemies).





Team Member:

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